

FemTech

1672023OWIDPSEn=166CPRn=138

log(MMR) -+ logGDP CPR -+ logGDP HC3OLS

0/1/2/3

CPR2010-20232015-2023

CPRHC3

OWID 2026-08-05V-Dem v16WDI JSONSHA-256

FemTech

f		URL		SDS-256				
Maternal mortality ratio	World Bank, World Development Indicators - APT suspended, last updated 2020-07-13	2020-09-03	https://api.worldbank.org/v2/country/all/indicator/SH.UA.MMRT?format=json&page=2000	All economies, latest non-missing observation in 2010-2023 for the retained 167-country analytic sample	Maternal deaths per 100,000 live births	Latest non-missing value, 2010-2023	mmr.json	677628321c88f2010813c5d0f93d6221832acc5d5f6153dc3d63e06d0d7c73
GDP per capita, PPP	World Bank, World Development Indicators - APT suspended, last updated 2020-07-13	2020-09-03	https://api.worldbank.org/v2/country/all/indicator/NY.GDP.PCAP.PP.KD?format=json&page=2	All economies, latest non-missing observation in 2010-2023 for the retained 167-country analytic sample	Constant 2021 international dollars per person	Latest non-missing value, 2010-2023, lag generated as UNData GDP	gdp.json	c13b48329239d4b79dadc38310f6c708bc3c4f5d45d31d8ac772d736a5
Contemporary prevalence, any method	World Bank, World Development Indicators - APT suspended, last updated 2020-07-13	2020-09-03	https://api.worldbank.org/v2/country/all/indicator/SP.DYN.CONU.PN?format=json&page=200	All economies, latest non-missing observation in 2010-2023 for the retained 167-country analytic sample	% of married women ages 15-49	Latest non-missing value, 2010-2023, missing retained and flagged	cpa.json	20823709d808086a9c93317464038fac9219823c302a067e0f2bc70a5
Political regime (Regions of the World)	V-Dem v10, processed by Our World in Data - COVID archive 2020-08-07 14:03 UTC, source last updated 2020-05-17	2020-09-03	https://archive.ourworldindata.org/20200807-143540-vragober/political-regime.html	2023 classification for the retained 167-country analytic sample	0 closed autocracy 1 electoral autocracy 2 electoral democracy 3 liberal democracy	Stand at 2023	1210103.data.json + 1210103.meta.data.json	050d9f752ac776d37b3d5d09bae470aef42235ac4534d74d43d73737e083d61558b4d6c3ba70746d783c32a57d4c39235d46ae179d80e0db7ca80222d6d
SDS-256	URL	WDS	JSON	OWID				

ISO3		MMR	MMR	GDP	GDP	logGDP		CPR	CPR	CPR	log(MMR)	Regime=1	Regime=2	Regime=3	CPR	-2018
AFG	Afghanistan	521.0	2023	1,983.81	2023	7.592776	0 Closed autocracy	2023	22.50	2015 Observed	6.255750	0	0	0	0	-3
ALB	Albania	7.0	2023	20,501.65	2023	9.928261	2 Electoral democracy	2023	46.00	2018 Observed	1.945910	0	1	0	0	0
DZA	Algeria	62.0	2023	15,159.32	2023	9.626371	1 Electoral autocracy	2023	53.60	2019 Observed	4.127134	1	0	0	0	1
AGO	Angola	183.0	2023	8,597.54	2023	9.059231	1 Electoral autocracy	2023	13.70	2016 Observed	5.209486	1	0	0	0	-2
ARG	Argentina	33.0	2023	27,230.40	2023	10.212089	2 Electoral democracy	2023	70.10	2020 Observed	3.496508	0	1	0	0	2
ARM	Armenia	19.0	2023	19,402.75	2023	9.873170	2 Electoral democracy	2023	57.10	2016 Observed	2.944439	0	1	0	0	-2
AUS	Australia	2.0	2023	60,684.96	2023	11.013451	3 Liberal democracy	2023	66.93	2016 Observed	0.093147	0	0	1	0	-2
AUT	Austria	6.0	2023	64,536.21	2023	11.071082	3 Liberal democracy	2023	65.65	2013 Observed	1.791759	0	0	1	0	-5
AZE	Azerbaijan	18.0	2023	21,310.74	2023	9.960966	1 Electoral autocracy	2023	54.90	2011 Observed	2.890372	1	0	0	0	-7
BHR	Bahrain	17.0	2023	57,819.91	2023	10.965088	0 Closed autocracy	2023		Missing	2.833213	0	0	0	0	
BGD	Bangladesh	115.0	2023	8,242.40	2023	9.017047	1 Electoral autocracy	2023	64.00	2022 Observed	4.749332	1	0	0	0	4
BRB	Barbados	35.0	2023	20,865.02	2023	9.945829	3 Liberal democracy	2023	59.21	2012 Observed	3.555348	0	0	1	0	-6
BLR	Belarus	1.0	2023	27,782.44	2023	10.232159	1 Electoral autocracy	2023	52.60	2019 Observed	0.000000	1	0	0	0	1
BEL	Belgium	4.0	2023	63,042.01	2023	11.051557	3 Liberal democracy	2023	66.70	2018 Observed	1.386294	0	0	1	0	0
BEN	Benin	518.0	2023	5,721.21	2023	8.221805	1 Electoral autocracy	2023	22.40	2022 Observed	6.240975	1	0	0	0	4
BTN	Bhutan	47.0	2023	14,610.30	2023	9.590482	3 Liberal democracy	2023	65.60	2010 Observed	3.850148	0	0	1	0	-4
BOL	Bolivia	146.0	2023	11,616.16	2023	9.360152	2 Electoral democracy	2023	66.50	2016 Observed	4.983607	0	1	0	0	-2
BIH	Bosnia and Herzegovina	6.0	2023	19,804.92	2023	8.893686	2 Electoral democracy	2023	46.02	2012 Observed	1.791759	0	1	0	0	-6
BWA	Botswana	155.0	2023	18,932.48	2023	9.848634	2 Electoral democracy	2023	67.40	2017 Observed	5.043425	0	1	0	0	-1
BRA	Brazil	67.0	2023	19,079.81	2023	9.856386	2 Electoral democracy	2023	80.20	2013 Observed	4.204693	0	1	0	0	-5
BGR	Bulgaria	6.0	2023	33,073.68	2023	10.406493	2 Electoral democracy	2023		Missing	1.791759	0	1	0	0	
BFA	Burkina Faso	242.0	2023	2,486.21	2023	7.818513	0 Closed autocracy	2023	33.96	2022 Observed	5.488938	0	0	0	0	4
BDI	Burundi	392.0	2023	1,033.08	2023	6.940272	1 Electoral autocracy	2023	28.50	2017 Observed	5.971262	1	0	0	0	-1
CPV	Cabo Verde	40.0	2023	9,238.46	2023	9.130048	2 Electoral democracy	2023	55.80	2018 Observed	3.688979	0	1	0	0	0
KHM	Cambodia	137.0	2023	6,695.29	2023	8.809160	1 Electoral autocracy	2023	61.90	2022 Observed	4.919081	1	0	0	0	4
CMR	Cameroon	258.0	2023	4,875.25	2023	8.491927	1 Electoral autocracy	2023	19.30	2018 Observed	5.552960	1	0	0	0	0
CAN	Canada	12.0	2023	58,088.38	2023	10.969721	2 Electoral democracy	2023		Missing	2.484907	0	1	0	0	
CAF	Central African Republic	692.0	2023	1,132.68	2023	7.032339	1 Electoral autocracy	2023	17.80	2019 Observed	6.539586	1	0	0	0	1
TCO	Chad	748.0	2023	2,433.53	2023	7.797100	0 Closed autocracy	2023	8.10	2019 Observed	6.617403	0	0	0	0	1
CHL	Chile	10.0	2023	29,600.42	2023	10.295544	3 Liberal democracy	2023	76.26	2016 Observed	2.302585	0	0	1	0	-2
CHN	China	16.0	2023	22,687.25	2023	10.629558	0 Closed autocracy	2023	84.50	2017 Observed	2.772899	0	0	0	0	-1
COL	Colombia	59.0	2023	18,404.62	2023	9.820466	2 Electoral democracy	2023	81.00	2013 Observed	4.077337	0	1	0	0	-3
COM	Comoros	179.0	2023	3,508.16	2023	8.162847	1 Electoral autocracy	2023	11.60	2022 Observed	5.187386	1	0	0	0	4
COD	Congo, Dem. Rep.	427.0	2023	1,559.34	2023	7.352017	1 Electoral autocracy	2023	28.13	2018 Observed	6.056784	1	0	0	0	0
COG	Congo, Rep.	241.0	2023	6,172.29	2023	8.727825	1 Electoral autocracy	2023	30.12	2015 Observed	5.487797	1	0	0	0	-3
CRI	Costa Rica	24.0	2023	26,378.54	2023	10.180306	3 Liberal democracy	2023	70.90	2018 Observed	3.178054	0	0	1	0	0
CTV	Cote d'Ivoire	359.0	2023	6,520.90	2023	8.782768	1 Electoral autocracy	2023	31.97	2022 Observed	5.883322	1	0	0	0	4
HRV	Croatia	3.0	2023	41,320.26	2023	10.629108	2 Electoral democracy	2023		Missing	1.098612	0	1	0	0	
CYP	Cyprus	14.0	2023	51,674.90	2023	10.852727	3 Liberal democracy	2023		Missing	2.639657	0	0	1	0	
CZE	Czechia	3.0	2023	47,558.41	2023	10.769714	3 Liberal democracy	2023		Missing	1.098612	0	0	1	0	
DNK	Denmark	4.0	2023	68,993.90	2023	11.141773	3 Liberal democracy	2023		Missing	1.386094	0	0	1	0	
DJI	Djibouti	162.0	2023	6,510.68	2023	8.781199	1 Electoral autocracy	2023	19.00	2012 Observed	5.087596	1	0	0	0	-6
DOM	Dominican Republic	124.0	2023	23,282.49	2023	10.055457	2 Electoral democracy	2023	62.78	2019 Observed	4.820282	0	1	0	0	1
ECU	Ecuador	55.0	2023	14,321.16	2023	9.569494	2 Electoral democracy	2023	77.85	2018 Observed	4.007333	0	1	0	0	0
EGY	Egypt, Arab Rep.	17.0	2023	16,691.39	2023	9.722648	1 Electoral autocracy	2023	58.50	2014 Observed	2.833213	1	0	0	0	-4
SLV	El Salvador	39.0	2023	11,404.33	2023	9.341749	1 Electoral autocracy	2023	53.60	2021 Observed	3.663562	1	0	0	0	3
GNQ	Equatorial Guinea	174.0	2023	15,802.15	2023	8.657749	1 Electoral autocracy	2023	12.60	2011 Observed	5.139055	1	0	0	0	-7
EST	Estonia	5.0	2023	41,284.77	2023	10.628249	3 Liberal democracy	2023	75.10	2014 Observed	1.609438	0	0	1	0	-4
SWZ	Eswatini	118.0	2023	10,181.50	2023	9.228328	0 Closed autocracy	2023	57.70	2022 Observed	4.779645	0	0	0	0	4
ETH	Ethiopia	195.0	2023	2,757.96	2023	7.922246	1 Electoral autocracy	2023	36.07	2021 Observed	5.273000	1	0	0	0	3
FJI	Fiji	30.0	2023	13,194.80	2023	9.487578	2 Electoral democracy	2023	35.40	2021 Observed	3.411197	0	1	0	0	3
FIN	Finland	8.0	2023	56,051.77	2023	10.934031	3 Liberal democracy	2023	85.47	2015 Observed	2.079442	0	0	1	0	-3
FRA	France	7.0	2023	54,296.72	2023	10.902219	3 Liberal democracy	2023	78.40	2010 Observed	1.945910	0	0	1	0	-8
GAB	Gabon	233.0	2023	18,701.42	2023	9.836355	0 Closed autocracy	2023	24.80	2021 Observed	5.451038	0	0	0	0	3
GMB	Gambia, The	354.0	2023	2,963.60	2023	7.994160	2 Electoral democracy	2023	18.90	2020 Observed	5.869297	0	1	0	0	2
GEO	Georgia	20.0	2023	22,590.53	2023	10.052386	2 Electoral democracy	2023	39.26	2018 Observed	2.995732	0	1	0	0	0
DEU	Germany	4.0	2023	63,140.19	2023	11.053113	3 Liberal democracy	2023	80.30	2011 Observed	1.386094	0	0	1	0	-7
GHA	Ghana	234.0	2023	6,808.54	2023	8.825933	2 Electoral democracy	2023	36.30	2022 Observed	5.455321	0	1	0	0	4
GRC	Greece	5.0	2023	36,700.51	2023	10.510546	2 Electoral democracy	2023		Missing	1.609438	0	1	0	0	
GTM	Guatemala	94.0	2023	12,380.37	2023	9.423868	2 Electoral democracy	2023	66.60	2015 Observed	4.543295	0	1	0	0	-3
GIN	Guinea	494.0	2023	3,904.82	2023	8.269968	0 Closed autocracy	2023	10.90	2018 Observed	6.202536	0	0	0	0	0
GNB	Guinea-Bissau	505.0	2023	2,660.81	2023	7.886388	1 Electoral autocracy	2023	21.20	2019 Observed	6.224558	1	0	0	0	1
GYF	Guyana	75.0	2023	49,512.36	2023	10.805990	1 Electoral autocracy	2023	29.88	2020 Observed	4.317488	0	1	0	0	2
HTI	Haiti	338.0	2023	2,939.12	2023	7.983867	0 Closed autocracy	2023	34.30	2017 Observed	5.793014	0	0	0	0	-1
HND	Honduras	47.0	2023	6,467.88	2023	8.774604	2 Electoral democracy	2023	69.37	2019 Observed	3.850148	0	1	0	0	1
HUN	Hungary	12.0	2023	40,333.25	2023	10.604932	1 Electoral autocracy	2023		Missing	2.484907	1	0	0	0	
ISL	Iceland	3.0	2023	69,268.12	2023	11.145740	3 Liberal democracy	2023		Missing	1.098612	0	0	1	0	
IND	India	80.0	2023	8,870.55	2023	9.090492	1 Electoral autocracy	2023	66.70	2021 Observed	4.382027	1	0	0	0	3
IDN	Indonesia	140.0	2023	13,889.95	2023	9.538921	2 Electoral democracy	2023	52.60	2023 Observed	4.941642	0	1	0	0	5
IRN	Iran, Islamic Rep.	16.0	2023	17,044.82	2023	9.743601	0 Closed autocracy	2023	77.40	2010 Observed	2.772589	0	0	0	0	-8
IRQ	Iraq	66.0	2023	13,802.52	2023	8.648148	1 Electoral autocracy	2023	52.79	2018 Observed	4.189655	1	0	0	0	0
IRL	Ireland	4.0	2023	117,659.78	2023	11.075352	3 Liberal democracy	2023	73.26	2010 Observed	1.386294	0	0	1	0	-8
ISR	Israel	2.0	2023	47,671.25	2023	10.772084	2 Electoral democracy	2023		Missing	0.093147	0	1	0	0	
ITA	Italy	6.0	2023	52,842.63	2023	10.875074	3 Liberal democracy	2023	65.10	2013 Observed	1.791759	0	0	1	0	-5
JAM	Jamaica	130.0	2023	11,398.81	2023	9.341264	2 Electoral democracy	2023	60.60	2022 Observed	4.867534	0	1	0	0	4
JPN	Japan	3.0	2023	47,387.22	2023	10.766108	3 Liberal democracy	2023	58.34	2021 Observed	1.098612	0	0	1	0	3
JOR	Jordan	31.0	2023	10,308.78	2023	9.240751	0 Closed autocracy	2023	60.10	2023 Observed	3.433987	0	0	0	0	5
KAZ	Kazakhstan	10.0	2023	34,703.23	2023	10.454588	1 Electoral autocracy	2023	52.98	2018 Observed	2.302585	1	0	0	0	0
KEN	Kenya	149.0	2023	5,691.83	2023	8.646786	2 Electoral democracy	2023	62.50	2022 Observed	5.803946	0	1	0	0	4
KOR	Korea, Rep.	4.0	2023	54,629.34	2023	10.897283	3 Liberal democracy									

LYB	Libya	59.0	2023	12,476.60	2023	9.431610	0 Closed autocracy	2023	27.70	2014 Observed	4.077537	0	0	0	-4
LTU	Lithuania	8.0	2023	46,533.90	2023	10.743629	2 Electoral democracy	2023		Missing	2.079442	0	1	0	
LUX	Luxembourg	11.0	2023	130,048.92	2023	11.775666	3 Liberal democracy	2023		Missing	2.397895	0	0	1	
MDG	Madagascar	445.0	2023	2,629.61	2023	7.396093	0 Closed autocracy	2023	49.70	2023 Observed	6.908074	1	0	0	3
MDI	Malawi	225.0	2023	1,649.30	2023	7.408105	2 Electoral democracy	2023	65.61	2020 Observed	5.416100	0	1	0	2
MYS	Malaysia	26.0	2023	32,857.67	2023	10.399940	2 Electoral democracy	2023	42.80	2022 Observed	3.258097	0	1	0	4
MDV	Maldives	32.0	2023	22,333.05	2023	10.013823	2 Electoral democracy	2023	18.80	2017 Observed	3.465736	0	1	0	-1
MLI	Mali	367.0	2023	2,861.48	2023	7.959094	0 Closed autocracy	2023	17.20	2018 Observed	5.905362	0	0	0	0
MLT	Malta	8.0	2023	60,459.77	2023	11.009733	2 Electoral democracy	2023		Missing	2.079442	0	1	0	
MRT	Mauritania	381.0	2023	6,276.48	2023	8.744564	1 Electoral autocracy	2023	14.30	2021 Observed	5.942799	1	0	0	3
MUS	Mauritius	66.0	2023	26,471.04	2023	10.183806	1 Electoral autocracy	2023	63.80	2014 Observed	4.189655	1	0	0	-4
MEX	Mexico	42.0	2023	21,864.92	2023	9.992619	2 Electoral democracy	2023	71.85	2023 Observed	3.376170	0	1	0	5
MDA	Moldova	19.0	2023	15,980.57	2023	9.679129	2 Electoral democracy	2023	56.00	2020 Observed	2.944439	0	1	0	2
MNG	Mongolia	41.0	2023	16,222.86	2023	9.694177	2 Electoral democracy	2023	48.13	2018 Observed	3.713572	0	1	0	0
MNE	Montenegro	6.0	2023	27,243.40	2023	10.212567	2 Electoral democracy	2023	20.70	2018 Observed	1.791759	0	1	0	0
MAR	Morocco	70.0	2023	8,915.35	2023	9.095530	0 Closed autocracy	2023	70.80	2018 Observed	4.248495	0	0	0	0
MOZ	Mozambique	99.0	2023	1,512.67	2023	7.321630	1 Electoral autocracy	2023	26.40	2023 Observed	4.595120	1	0	0	5
MMR	Myanmar	185.0	2023	5,364.14	2023	8.587491	0 Closed autocracy	2023	52.20	2016 Observed	5.220556	0	0	0	-2
NAM	Namibia	139.0	2023	10,126.24	2023	9.222885	2 Electoral democracy	2023	56.10	2013 Observed	4.994474	0	1	0	-5
NPL	Nepal	142.0	2023	4,861.23	2023	8.489047	2 Electoral democracy	2023	57.20	2022 Observed	4.955827	0	1	0	4
NLD	Netherlands	4.0	2023	70,198.95	2023	11.159089	3 Liberal democracy	2023	73.00	2013 Observed	1.386294	0	0	1	-5
NZL	New Zealand	7.0	2023	49,592.59	2023	10.811597	3 Liberal democracy	2023		Missing	1.945910	0	0	1	
NIC	Nicaragua	60.0	2023	7,496.53	2023	8.922196	1 Electoral autocracy	2023	80.40	2012 Observed	4.094345	1	0	0	-6
NER	Niger	350.0	2023	1,712.97	2023	7.445984	0 Closed autocracy	2023	15.26	2023 Observed	5.857933	0	0	0	5
NGA	Nigeria	993.0	2023	7,843.00	2023	8.963737	1 Electoral autocracy	2023	21.60	2021 Observed	6.900731	1	0	0	3
MKD	North Macedonia	3.0	2023	23,452.22	2023	10.062848	2 Electoral democracy	2023	59.87	2019 Observed	1.098612	0	1	0	1
NOR	Norway	1.0	2023	94,364.96	2023	11.454925	3 Liberal democracy	2023		Missing	0.000000	0	0	1	
OMN	Oman	13.0	2023	37,793.62	2023	10.530995	0 Closed autocracy	2023	29.70	2014 Observed	2.564949	0	0	0	-4
PAK	Pakistan	155.0	2023	5,418.77	2023	8.597624	1 Electoral autocracy	2023	34.00	2019 Observed	5.043425	1	0	0	1
PAN	Panama	37.0	2023	35,872.97	2023	10.487739	2 Electoral democracy	2023	50.80	2015 Observed	3.610918	0	1	0	-3
PNG	Papua New Guinea	189.0	2023	4,206.38	2023	8.344357	1 Electoral autocracy	2023	36.70	2018 Observed	5.241747	1	0	0	0
PRY	Paraguay	58.0	2023	15,831.58	2023	9.669762	2 Electoral democracy	2023	68.36	2016 Observed	4.060443	0	1	0	-2
PER	Peru	51.0	2023	15,342.27	2023	9.638367	2 Electoral democracy	2023	77.30	2023 Observed	3.931826	0	1	0	5
PHL	Philippines	84.0	2023	9,898.53	2023	9.200142	1 Electoral democracy	2023	58.30	2022 Observed	4.430817	1	0	0	4
POL	Poland	2.0	2023	45,672.77	2023	10.684480	2 Electoral democracy	2023	63.20	2014 Observed	0.693147	0	1	0	-4
PRT	Portugal	15.0	2023	41,764.26	2023	10.698092	2 Electoral democracy	2023	73.89	2014 Observed	2.708050	0	1	0	-4
QAT	Qatar	4.0	2023	116,159.14	2023	11.662716	0 Closed autocracy	2023	37.52	2012 Observed	1.386294	0	0	0	-6
ROU	Romania	12.0	2023	40,116.69	2023	10.599548	2 Electoral democracy	2023		Missing	2.484907	0	1	0	
RUS	Russian Federation	9.0	2023	39,881.45	2023	10.593667	1 Electoral autocracy	2023	68.00	2011 Observed	2.197225	1	0	0	-7
RWA	Rwanda	229.0	2023	3,168.03	2023	8.060865	1 Electoral autocracy	2023	64.10	2020 Observed	5.433722	1	0	0	2
STP	Sao Tome and Principe	75.0	2023	5,541.30	2023	8.619985	2 Electoral democracy	2023	49.68	2019 Observed	4.317488	0	1	0	1
SAU	Saudi Arabia	7.0	2023	64,481.84	2023	11.074139	0 Closed autocracy	2023	39.99	2023 Observed	1.945910	0	0	0	5
SEN	Senegal	237.0	2023	4,304.76	2023	8.367477	2 Electoral democracy	2023	26.50	2023 Observed	5.468060	0	1	0	5
SRB	Serbia	11.0	2023	25,735.13	2023	10.156142	1 Electoral autocracy	2023	62.30	2019 Observed	2.397895	1	0	0	1
SYC	Seychelles	42.0	2023	28,713.67	2023	10.265128	3 Liberal democracy	2023		Missing	3.375670	0	0	1	
SLE	Sierra Leone	354.0	2023	3,034.95	2023	8.017949	1 Electoral autocracy	2023	21.20	2019 Observed	5.862997	1	0	0	1
SGP	Singapore	6.0	2023	130,296.84	2023	11.777371	1 Electoral autocracy	2023		Missing	1.791759	1	0	0	
SVK	Slovak Republic	4.0	2023	39,501.95	2023	10.584105	2 Electoral democracy	2023		Missing	1.386294	0	1	0	
SVN	Slovenia	3.0	2023	47,986.42	2023	10.778673	2 Electoral democracy	2023		Missing	1.098612	0	1	0	
SLB	Solomon Islands	123.0	2023	2,339.87	2023	7.757853	2 Electoral democracy	2023	29.30	2015 Observed	4.812184	0	1	0	-3
SOM	Somalia, Fed. Rep.	563.0	2023	1,401.65	2023	7.245404	0 Closed autocracy	2023	6.90	2019 Observed	6.333280	0	0	0	1
ZAF	South Africa	118.0	2023	13,695.38	2023	9.531814	2 Electoral democracy	2023	54.60	2018 Observed	4.770685	0	1	0	-2
ESP	Spain	3.0	2023	47,322.59	2023	10.764743	3 Liberal democracy	2023	62.08	2018 Observed	1.098612	0	0	1	0
LKA	Sri Lanka	18.0	2023	13,065.07	2023	9.477698	2 Electoral democracy	2023	64.60	2016 Observed	2.890372	0	1	0	-2
SDN	Sudan	256.0	2023	2,181.08	2023	7.687375	0 Closed autocracy	2023	12.78	2014 Observed	5.545177	0	0	0	-4
SUR	Suriname	84.0	2023	19,020.87	2023	9.853292	3 Liberal democracy	2023	39.12	2018 Observed	4.430817	0	0	1	0
SWE	Sweden	4.0	2023	62,122.51	2023	11.036864	3 Liberal democracy	2023		Missing	1.386294	0	0	1	
CHE	Switzerland	5.0	2023	85,383.36	2023	11.354906	3 Liberal democracy	2023	71.60	2017 Observed	1.609438	0	0	1	-1
SYR	Syrian Arab Republic	20.0	2023	4,579.44	2022	8.429332	0 Closed autocracy	2023		Missing	2.995732	0	0	0	
TJK	Tajikistan	14.0	2023	4,472.32	2023	8.405645	1 Electoral autocracy	2023	29.30	2017 Observed	2.539057	1	0	0	-1
TZA	Tanzania	276.0	2023	3,620.79	2023	8.194448	1 Electoral autocracy	2023	37.60	2022 Observed	5.620401	1	0	0	4
THA	Thailand	34.0	2023	21,245.27	2023	9.963890	1 Electoral autocracy	2023	72.80	2022 Observed	3.526361	1	0	0	4
TLS	Timor-Leste	192.0	2023	4,331.08	2023	8.373372	2 Electoral democracy	2023	26.00	2016 Observed	5.257495	0	1	0	-2
TGO	Togo	349.0	2023	3,215.82	2023	8.075839	1 Electoral autocracy	2023	23.80	2017 Observed	5.855072	1	0	0	-1
TTO	Trinidad and Tobago	54.0	2023	31,196.60	2023	10.348065	3 Liberal democracy	2023	36.50	2022 Observed	3.988984	0	0	1	4
TUN	Tunisia	36.0	2023	12,650.96	2023	9.445489	1 Electoral autocracy	2023	54.00	2023 Observed	3.583519	1	0	0	5
TUR	Turkey	15.0	2023	35,069.17	2023	10.465078	1 Electoral autocracy	2023	69.80	2018 Observed	2.708050	1	0	0	0
TKM	Turkmenistan	5.0	2023	17,685.86	2023	9.786518	0 Closed autocracy	2023	49.68	2019 Observed	1.609438	0	0	0	1
UGA	Uganda	170.0	2023	2,791.06	2023	7.934175	1 Electoral autocracy	2023	43.00	2022 Observed	5.135798	1	0	0	4
UKR	Ukraine	15.0	2023	15,916.96	2023	9.675140	1 Electoral autocracy	2023	65.46	2012 Observed	2.708050	1	0	0	-6
ARE	United Arab Emirates	3.0	2023	70,239.83	2023	11.159671	0 Closed autocracy	2023		Missing	1.098612	0	0	0	
GBR	United Kingdom	8.0	2023	53,423.42	2023	10.886005	3 Liberal democracy	2023	76.10	2012 Observed	2.079442	0	0	1	-6
USA	United States	17.0	2023	74,351.67	2023	11.216561	3 Liberal democracy	2023	73.90	2019 Observed	2.833213	0	0	1	1
URY	Uruguay	15.0	2023	31,100.97	2023	10.344994	3 Liberal democracy	2023		Missing	2.708050	0	0	1	
UZB	Uzbekistan	26.0	2023	10,514.10	2023	9.260472	0 Closed autocracy	2023	58.80	2022 Observed	3.258097	0	0	0	4
VUT	Vanuatu	100.0	2023	3,762.83	2023	8.230227	2 Electoral democracy	2023	49.80	2013 Observed	4.005170	0	1	0	-5
VNM	Viet Nam	48.0	2023	13,534.80	2023	9.513019	0 Closed autocracy	2023	72.80	2021 Observed	3.871201	0	0	0	5
PSE	West Bank and Gaza	16.0	2023	5,562.42	2023	8.623789	Not available		57.30	2020 Observed	2.727589	0	0	0	2 OWID 2023 reports Palestine/Gaza and Palestine/West Bank separately; no unified PSE regime value.
ZMB	Zambia	85.0	2023	3,673.48	2023	8.208896	2 Electoral democracy	2023	49.60	2018 Observed	4.426551	0	1	0	0
ZWE	Zimbabwe	358.0	2023	5,218.05	2023	8.559878	1 Electoral autocracy	2023	66.80	2015 Observed	5.880533	1	0	0	-3

“ ”		QUARTILE.I NC SD					

	n	SD		IQR			
MMR per 100,000 live births	167	116.940	#NAME?	41.000	#NAME?	1.000	993.000 mmr
log(MMR)	167	3.637	#NAME?	3.714	#NAME?	0.000	6.901 log_mmr
GDP per capita, PPP, constant 2021 intl \$	167	25 423.047	#NAME?	15 916.958	#NAME?	1033.009	gdp_ppp_constant_2021
log(GDP per capita)	167	9.571	#NAME?	9.675	#NAME?	6.940	11.778 log_gdp
Political regime code (0-3)	166	1.566	#NAME?	2.000	#NAME?	0.000	3.000 regime_code
Contraceptive prevalence, any method (%)	139	49.896	#NAME?	54.600	#NAME?	6.900	85.475 cpr
CPR observation year	139	2017.741	#NAME?	2018.000	#NAME?	2010.000	2023.000 cpr_year

CPR
PSE

	CPR	CPR		
0 Closed autocracy	27	24	3	11.1%
1 Electoral autocracy	51	47	4	7.8%
2 Electoral democracy	55	45	10	18.2%
3 Liberal democracy	33	22	11	33.3%

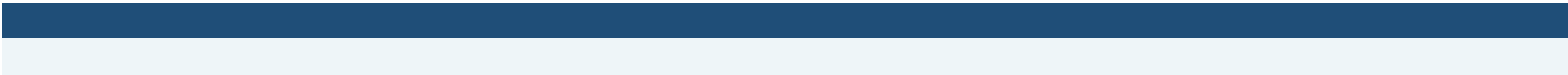
CPR			
ISO3			
BHR	Bahrain	0 Closed autocracy	
BGR	Bulgaria	2 Electoral democracy	
CAN	Canada	2 Electoral democracy	
HRV	Croatia	2 Electoral democracy	
CYP	Cyprus	3 Liberal democracy	CPR
CZE	Czechia	3 Liberal democracy	CPR
DNK	Denmark	3 Liberal democracy	CPR
GRC	Greece	2 Electoral democracy	
HUN	Hungary	1 Electoral autocracy	
ISL	Iceland	3 Liberal democracy	CPR
ISR	Israel	2 Electoral democracy	
KWT	Kuwait	1 Electoral autocracy	
LVA	Latvia	3 Liberal democracy	CPR
LBN	Lebanon	1 Electoral autocracy	
LTU	Lithuania	2 Electoral democracy	
LUX	Luxembourg	3 Liberal democracy	CPR
MLT	Malta	2 Electoral democracy	
NZL	New Zealand	3 Liberal democracy	CPR
NOR	Norway	3 Liberal democracy	CPR
ROU	Romania	2 Electoral democracy	
SYC	Seychelles	3 Liberal democracy	CPR
SGP	Singapore	1 Electoral autocracy	
SVK	Slovak Republic	2 Electoral democracy	
SVN	Slovenia	2 Electoral democracy	
SWE	Sweden	3 Liberal democracy	CPR
SYR	Syrian Arab Republic	0 Closed autocracy	
ARE	United Arab Emirates	0 Closed autocracy	
URY	Uruguay	3 Liberal democracy	CPR

Pearson r MMR CPR n=167

X	Y	n	Pearson r
	log(MMR)	166	-0.441
	MMR	166	-0.368
	CPR	138	0.446
logGDP	log(MMR)	167	-0.850
logGDP	MMR	167	-0.716
logGDP	CPR	139	0.575
	logGDP	166	0.489

HC3 OLS β 0-3

		n	df	β	SE	p	HC3 SE	HC3 p	HC3 95%CI	HC3 95%CI	R ²	R ²
M1 Primary: log(MMR), ordinal regime	regime_code	166	163	-0.0531	0.0792	0.5038	0.0752	0.4812	-0.2015	0.0954	0.7312	0.7279
M3 Primary: CPR 2010-2023 latest	regime_code	138	135	4.8273	1.6771	0.0047	1.9817	0.0162	0.9081	8.7465	0.3726	0.3633
logGDP												
		n	df	β	SE	p	HC3 SE	HC3 p	HC3 95%CI	HC3 95%CI	R ²	R ²
M1 Primary: log(MMR), ordinal regime	Intercept	166	163	15.2500	0.5924	<0.0001	0.5286	<0.0001	14.2063	16.2938	0.7312	0.7279
M1 Primary: log(MMR), ordinal regime	regime_code	166	163	-0.0531	0.0792	0.5038	0.0752	0.4812	-0.2015	0.0954	0.7312	0.7279
M1 Primary: log(MMR), ordinal regime	log_gdp	166	163	-1.2034	0.0667	<0.0001	0.0611	<0.0001	-1.3241	-1.0826	0.7312	0.7279
M3 Primary: CPR 2010-2023 latest	Intercept	138	135	-39.6847	12.5798	0.0020	12.8337	0.0024	-65.0659	-14.3035	0.3726	0.3633
M3 Primary: CPR 2010-2023 latest	regime_code	138	135	4.8273	1.6771	0.0047	1.9817	0.0162	0.9081	8.7465	0.3726	0.3633
M3 Primary: CPR 2010-2023 latest	log_gdp	138	135	8.8203	1.4418	<0.0001	1.5844	<0.0001	5.6867	11.9538	0.3726	0.3633



		n	df	β	SE	p	HC3 SE	HC3 p	HC3 95%CI	HC3 95%CI	R ²	R ²
M2 Sensitivity: raw MMR, ordinal regime	regime_code	166	163	-3.4874	10.6188	0.7430	9.2455	0.7065	-21.7437	14.7690	0.5204	0.5145
M3 Primary: CPR 2010-2023 latest	regime_code	138	135	4.8273	1.6771	0.0047	1.9817	0.0162	0.9081	8.7465	0.3726	0.3633
M4 CPR + observation year	regime_code	138	134	4.8005	1.6868	0.0051	2.0299	0.0195	0.7856	8.8153	0.3729	0.3588
M5 CPR restricted to 2015-2023	regime_code	109	106	3.2653	2.0153	0.1081	2.3616	0.1697	-1.4169	7.9474	0.3420	0.3296
M6 log(MMR), regime categorical	regime_1	166	161	0.1026	0.2095	0.6252	0.2135	0.6317	-0.3191	0.5243	0.7322	0.7256
M6 log(MMR), regime categorical	regime_2	166	161	-0.0455	0.2092	0.8279	0.2035	0.8232	-0.4475	0.3564	0.7322	0.7256
M6 log(MMR), regime categorical	regime_3	166	161	-0.1190	0.2572	0.6442	0.2409	0.6219	-0.5947	0.3566	0.7322	0.7256
M7 CPR, regime categorical	regime_1	138	133	3.3902	4.1949	0.4204	4.9425	0.4939	-6.3859	13.1663	0.3757	0.3570
M7 CPR, regime categorical	regime_2	138	133	10.6131	4.2546	0.0138	5.1247	0.0403	0.4766	20.7495	0.3757	0.3570
M7 CPR, regime categorical	regime_3	138	133	12.4774	5.6393	0.0286	6.3825	0.0527	-0.1470	25.1018	0.3757	0.3570

OLS HC3			
	<i>p</i>	95%	<i>t</i>

				n	k	df	β	SE	t	p	CI	CI	HC3 SE	HC3 t	HC3 p	HC3 CI	HC3 CI	R ²	R ²
M1 Primary: log(MMR), ordinal regime	log_mmr	regime_code + log_gdp	Intercept	166	3	163	15.250020	0.592408	25.742423	<0.000001	14.080236	16.419804	0.528587	28.850531	<0.000001	14.206259	16.293781	0.731169	0.727871
M1 Primary: log(MMR), ordinal regime	log_mmr	regime_code + log_gdp	regime_code	166	3	163	-0.053072	0.079202	-0.670083	0.503753	-0.209465	0.103322	0.075181	-0.705918	0.481247	-0.201526	0.095383	0.731169	0.727871
M1 Primary: log(MMR), ordinal regime	log_mmr	regime_code + log_gdp	log_gdp	166	3	163	-1.203359	0.066730	-18.033237	<0.000001	-1.335126	-1.071592	0.061143	-19.681015	<0.000001	-1.324094	-1.082624	0.731169	0.727871
M2 Sensitivity: raw MMR, ordinal regime	mmr	regime_code + log_gdp	Intercept	166	3	163	1103.087721	79.425786	13.888282	<0.000001	946.251607	1259.923834	98.594433	11.188134	<0.000001	908.400725	1297.774716	0.520380	0.514495
M2 Sensitivity: raw MMR, ordinal regime	mmr	regime_code + log_gdp	regime_code	166	3	163	-3.487378	10.618782	-0.328416	0.743019	-24.455486	17.480731	9.245488	-0.377198	0.706517	-21.743746	14.768990	0.520380	0.514495
M2 Sensitivity: raw MMR, ordinal regime	mmr	regime_code + log_gdp	log_gdp	166	3	163	-102.336255	8.946685	-11.438455	<0.000001	-120.002599	-84.669910	10.158210	-10.074241	<0.000001	-122.394907	-82.277602	0.520380	0.514495
M3 Primary: CPR 2010-2023 latest	cpr	regime_code + log_gdp	Intercept	138	3	135	-39.684710	12.579757	-3.154648	0.001981	-64.563598	-14.805823	12.833748	-3.092215	0.002414	-65.065914	-14.303507	0.372602	0.363307
M3 Primary: CPR 2010-2023 latest	cpr	regime_code + log_gdp	regime_code	138	3	135	4.827299	1.677145	2.878283	0.004650	1.510422	8.144176	1.981713	2.435922	0.016158	0.908080	8.746518	0.372602	0.363307
M3 Primary: CPR 2010-2023 latest	cpr	regime_code + log_gdp	log_gdp	138	3	135	8.820277	1.441752	6.117749	<0.000001	5.968935	11.671618	1.584436	5.566823	<0.000001	5.686749	11.953804	0.372602	0.363307
M4 CPR + observation year	cpr	regime_code + log_gdp + cpr_year_centered_2018	Intercept	138	4	134	-38.760326	13.201521	-2.936050	0.003914	-64.870634	-12.650018	13.515270	-2.867891	0.004802	-65.491176	-12.029476	0.372870	0.358830
M4 CPR + observation year	cpr	regime_code + log_gdp + cpr_year_centered_2018	regime_code	138	4	134	4.800465	1.686762	2.845965	0.005124	1.464344	8.136587	2.029920	2.364854	0.019472	0.785637	8.815294	0.372870	0.358830
M4 CPR + observation year	cpr	regime_code + log_gdp + cpr_year_centered_2018	log_gdp	138	4	134	8.722590	1.503281	5.802367	<0.000001	5.749362	11.695819	1.631349	5.346856	<0.000001	5.496066	11.949115	0.372870	0.358830
M4 CPR + observation year	cpr	regime_code + log_gdp + cpr_year_centered_2018	cpr_year_centered_2018	138	4	134	-0.101571	0.424382	-0.239338	0.811209	-0.940923	0.737782	0.468550	-0.216777	0.828712	-1.028281	0.825139	0.372870	0.358830
M5 CPR restricted to 2015-2023	cpr	regime_code + log_gdp	Intercept	109	3	106	-44.284331	14.497330	-3.054654	0.002850	-73.026697	-15.541966	15.186082	-2.916113	0.004326	-74.392217	-14.176446	0.342040	0.329626
M5 CPR restricted to 2015-2023	cpr	regime_code + log_gdp	regime_code	109	3	106	3.265265	2.015267	1.620264	0.108147	-0.730199	7.260728	2.361611	1.382643	0.169680	-1.416860	7.947389	0.342040	0.329626
M5 CPR restricted to 2015-2023	cpr	regime_code + log_gdp	log_gdp	109	3	106	9.548834	1.697777	5.624317	<0.000001	6.182827	12.914841	1.892366	5.045976	0.000002	5.797034	13.300634	0.342040	0.329626
M6 log(MMR), regime categorical	log_mmr	regime_1 + regime_2 + regime_3 + log_gdp	Intercept	166	5	161	15.056338	0.666015	22.606608	<0.000001	13.741087	16.371590	0.622179	24.199387	<0.000001	13.827655	16.285021	0.732244	0.725592
M6 log(MMR), regime categorical	log_mmr	regime_1 + regime_2 + regime_3 + log_gdp	regime_1	166	5	161	0.102561	0.209532	0.489479	0.625169	-0.311224	0.516346	0.213536	0.480301	0.631665	-0.319130	0.524253	0.732244	0.725592
M6 log(MMR), regime categorical	log_mmr	regime_1 + regime_2 + regime_3 + log_gdp	regime_2	166	5	161	-0.045545	0.209176	-0.217736	0.827910	-0.458627	0.367537	0.203523	-0.223784	0.823209	-0.447465	0.356374	0.732244	0.725592
M6 log(MMR), regime categorical	log_mmr	regime_1 + regime_2 + regime_3 + log_gdp	regime_3	166	5	161	-0.119014	0.257247	-0.462646	0.644243	-0.627027	0.388999	0.240854	-0.494133	0.621886	-0.594655	0.356627	0.732244	0.725592
M6 log(MMR), regime categorical	log_mmr	regime_1 + regime_2 + regime_3 + log_gdp	log_gdp	166	5	161	-1.191059	0.070579	-16.875568	<0.000001	-1.330439	-1.051679	0.066984	-17.781117	<0.000001	-1.323341	-1.058778	0.732244	0.725592
M7 CPR, regime categorical	cpr	regime_1 + regime_2 + regime_3 + log_gdp	Intercept	138	5	133	-40.912470	14.435648	-2.834128	0.005312	-69.465623	-12.359317	14.728842	-2.777711	0.006266	-70.045548	-11.779392	0.375746	0.356972
M7 CPR, regime categorical	cpr	regime_1 + regime_2 + regime_3 + log_gdp	regime_1	138	5	133	3.390243	4.194921	0.808178	0.420431	-4.907148	11.687634	4.942514	0.685935	0.493948	-6.385858	13.166343	0.375746	0.356972
M7 CPR, regime categorical	cpr	regime_1 + regime_2 + regime_3 + log_gdp	regime_2	138	5	133	10.613061	4.254607	2.494487	0.013839	2.197612	19.028510	5.124689	2.070967	0.040295	0.476625	20.749497	0.375746	0.356972
M7 CPR, regime categorical	cpr	regime_1 + regime_2 + regime_3 + log_gdp	regime_3	138	5	133	12.477392	5.639327	2.212568	0.028633	1.323022	23.631762	6.382512	1.954934	0.052688	-0.146970	25.101754	0.375746	0.356972
M7 CPR, regime categorical	cpr	regime_1 + regime_2 + regime_3 + log_gdp	log_gdp	138	5	133	9.004782	1.559581	5.773849	<0.000001	5.919993	12.089572	1.696647	5.307400	<0.000001	5.648881	12.360684	0.375746	0.356972

HC3: $(XX)^{-1} - 1'X' \text{diag}[e^2/(1-h_i)^2] X (XX)^{-1}$. Two-sided $p = 2 \cdot \text{Pr}(T(df) \geq |t|)$, $df = n - k$, 95% $CI = \beta \pm 10.975/df \cdot SEHC3$.

Modits, "Privacy Not Included" review	20 prenatal/pregnancy apps and 1 wearable device	Mobile apps and connected wearables	Products available to US consumers	25	18	23 Products assigned Modits's "Privacy Not Included" warning	Desk-based product privacy/security review against Modits criteria	Independent civil society review	Includes 7 devices; warning criteria are not equivalent to observed third-party transmission	Product-level practices may have changed after 2021 (the cited page is a contemporaneous review)
									https://www.moditsfoundations.org/en/health-to-past-one-a-week-as-modits-labels-18-of-21-pregnancy-apps/	
Grandy et al., BMJ (2019)	24 medicine-related Android apps	Android	Apps available in four countries	26	19	14 Apps sharing user data outside the app ecosystem	Traffic analysis plus network sniffing of data recipients	Peer-reviewed empirical study	Not a representative health app sample; equivalent differs from Modits's warning label	No product-level consultation assessment in this workshop
									https://www.bmj.com/content/364/bmj.h109	
FTC Real order Re Modits (2021)	Five prenatal and evolution tracking app	Mobile app	United States enforcement			Alleged disclosure of sensitive health data to analytics/marketing firms contrary to privacy promises	Regulatory investigation and consent order	Individual enforcement case	A case, not a prevalence estimate; cannot be pooled with multi-proportion	It states that an independent privacy review required by the order was completed in March 2021.
									https://www.ftc.gov/news-events/news/press-releases/2021-04/ftc-modits-order-for-healthy-fertility	
FTC proposed order: Prenatal/Early Healthmate (2021)	Prenatal evolution tracking app	Mobile app	United States enforcement			Alleged sharing of sensitive health data, proposed order and US\$100,000 civil penalty	Regulatory investigation and proposed consent order	Individual enforcement case	A case, not a prevalence estimate; cannot be pooled with multi-proportion	The workshopper does not independently verify false compliance
									https://www.ftc.gov/news-events/news/press-releases/2021-03-15/ftc-proposed-order-tracking-apps-renewal	

PASS

=167	PASS	167
MMR n=167	PASS	MMR
n=166	PASS	PSE
	PASS	0/1/2/3
CPR n=139	PASS	CPR
CPR n=138	PASS	PSE 1
CPR =28	PASS	
CPR 2010-2023	PASS	
	PASS	

```

# Requires: Python 3, NumPy, and openpyxl. No dedicated statistical package is used.
# p values and 95% CIs use the t distribution with finite degrees of freedom (not a normal approximation).
#      5-171      C=MMR E= GDP H=      K=CPR L=CPR
#      G      logGDP = LN(E) N      log(MMR) = LN(C) O/Q/R
#      HC3      1e-10
import numpy as np, openpyxl
from math import lgamma, exp, log
#
import glob as _glob, os as _os
_cand = _glob.glob(_os.path.join(_os.path.dirname(_os.path.abspath(__file__))) or "", "** *xlsx")
if not _cand:
    raise SystemExit("      .xlsx ")
wb = openpyxl.load_workbook(_cand[0])
ws = wb["      "]
rows = [r for r in ws.iter_rows(min_row=5, max_row=171, values_only=True) if r and r[0]]
D = [dict(mmr=float(r[2]), gdp=float(r[4]), code=r[7], cpr=r[10], cyear=r[11]) for r in rows]
def fnt_p(p):
    #      p      <0.0001      0
    return "<0.0001" if p < 1e-4 else f"{p:.4f}"

def t_p(t, df):
    #      p      beta
    x = df / (df + t * t); a, b = df / 2, 0.5
    def bcf(a, b, x, itmax=300, eps=3e-14):
        qab, qap, qam = a + b, a + 1, a - 1
        c = 1.0; d = 1 - qab * x / qap; d = 1 / d; h = d
        for m in range(1, itmax + 1):
            m2 = 2 * m
            aa = m * (b - m) * x / ((qam + m2) * (a + m2))
            d = 1 + aa * d; c = 1 + aa / c; d = 1 / d; h *= d * c
            aa = -(a + m) * (qab + m) * x / ((a + m2) * (qap + m2))
            d = 1 + aa * d; c = 1 + aa / c; d = 1 / d; de = d * c; h *= de
            if abs(de - 1) < eps: break
        return h
    lb = lgamma(a) + lgamma(b) - lgamma(a + b)
    return (exp(a * log(x) + b * log(1 - x) - lb) * bcf(a, b, x) / a if x < (a + 1) / (a + b + 2)
           else 1 - exp(b * log(1 - x) + a * log(x) - lb) * bcf(b, a, 1 - x) / b)

def ols(y, Xcols, names):
    n = len(y); X = np.column_stack([np.ones(n)] + Xcols); k = X.shape[1]
    XtXi = np.linalg.inv(X.T @ X); beta = XtXi @ X.T @ y
    e = y - X @ beta; dof = n - k
    se_ols = np.sqrt(np.diag((e @ e / dof) * XtXi))
    h = np.diag(X @ XtXi @ X.T)

```

```

V = XtXi @ (X.T @ np.diag((e ** 2) / ((1 - h) ** 2)) @ X) @ XtXi    # HC3
se_hc3 = np.sqrt(np.diag(V))
r2 = 1 - (e @ e) / ((y - y.mean()) ** 2).sum()
for i, nm in enumerate(names):
    print(f"{nm:10s} beta={beta[i]:9.4f} SE={se_ols[i]:8.4f} p={fmt_p(t_p(abs(beta[i]/se_ols[i]), dof))>8} "
          f"| HC3 SE={se_hc3[i]:8.4f} p={fmt_p(t_p(abs(beta[i]/se_hc3[i]), dof))>8}")
    print(f"n={n} dof={dof} R2={r2:.4f} adjR2={1-(1-r2)*(n-1)/dof:.4f}")
lg = lambda g: np.log(g)
# M1 / M2                2023    PSE  n=166
D_c  = [d for d in D if d["code"] is not None]
code = np.array([d["code"] for d in D_c], dtype=float)
lmmr_c = np.array([np.log(d["mmr"]) for d in D_c])
mmr_c = np.array([d["mmr"] for d in D_c])
lgdp_c = np.array([lg(d["gdp"]) for d in D_c])
ols(lmmr_c, [code, lgdp_c], ["Intercept", "regime", "logGDP"])    # M1
ols(lmmr_c, [code, lgdp_c], ["Intercept", "regime", "logGDP"])    # M2
# M3 / M4 / M5  CPR      n=138  M5   2015-2023   n=109
sub  = [d for d in D if d["cpr"] is not None and d["code"] is not None]
cpr  = np.array([d["cpr"] for d in sub])
c_cd = np.array([d["code"] for d in sub], dtype=float)
c_lg = np.array([lg(d["gdp"]) for d in sub])
ols(cpr, [c_cd, c_lg], ["Intercept", "regime", "logGDP"])          # M3
ols(cpr, [c_cd, c_lg, np.array([d["cyear"] for d in sub], dtype=float) - 2018],
     ["Intercept", "regime", "logGDP", "cpr_year_c"])              # M4
sub15 = [d for d in sub if d["cyear"] >= 2015]
ols(np.array([d["cpr"] for d in sub15]),
     [np.array([d["code"] for d in sub15], dtype=float),
      np.array([lg(d["gdp"]) for d in sub15])],
     ["Intercept", "regime", "logGDP"])                            # M5
# M6 / M7                =
d1 = np.array([1.0 if d["code"] == 1 else 0.0 for d in D_c])
d2 = np.array([1.0 if d["code"] == 2 else 0.0 for d in D_c])
d3 = np.array([1.0 if d["code"] == 3 else 0.0 for d in D_c])
ols(lmmr_c, [d1, d2, d3, lgdp_c], ["Intercept", "regime_1", "regime_2", "regime_3", "logGDP"]) # M6
e1 = np.array([1.0 if d["code"] == 1 else 0.0 for d in sub])
e2 = np.array([1.0 if d["code"] == 2 else 0.0 for d in sub])
e3 = np.array([1.0 if d["code"] == 3 else 0.0 for d in sub])
ols(cpr, [e1, e2, e3, c_lg], ["Intercept", "regime_1", "regime_2", "regime_3", "logGDP"])    # M7

```